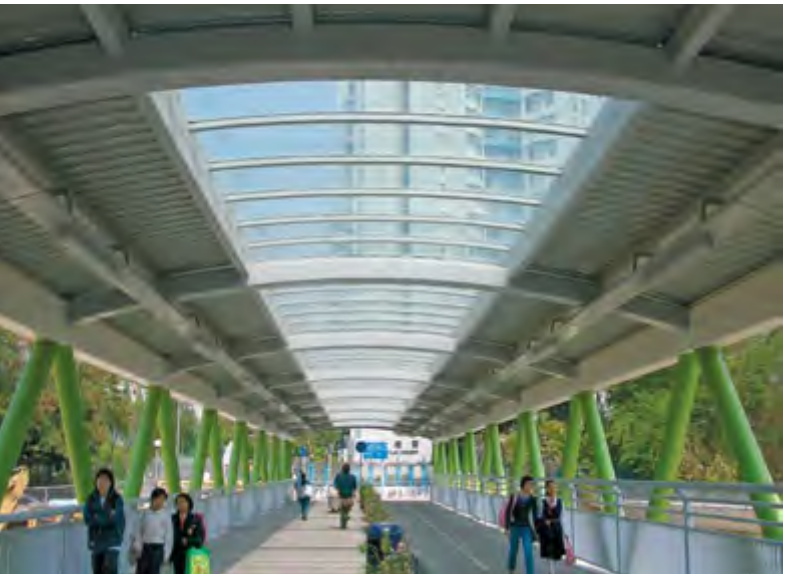
PALRAM

TufLite









# Palram India Product Range

## Wide Range Of Polycarbonate & PVC Roofing and Cladding

50 YEARS
Global Excellence
Since 1963

Scan for Video

www.palram.com





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# About Palram

## A Global Leader in Polycarbonate & PVC Panels

Palram India Pvt Ltd is a 100% subsidiary of Palram Industries Ltd , a multinational manufacturer with 8 strategic plants located around the world. Palram manufactures a wide range of Polycarbonate & PVC Panels for a wide variety of projects and applications including Roofing & Cladding , Architectural projects , signage and printing , Agriculture , fabrication and DIY . Palram's global presence and advanced technological abilities allow us to provide our customers with superior competitive products , while maintaining a high level of service standards. Palram delivers excellence to a global marketplace, backed by professional support and service on both local and regional levels. Palram is proud of its unique corporate culture that makes us agile, creative and committed to all our customers.

## Panel Extrusion Pioneering

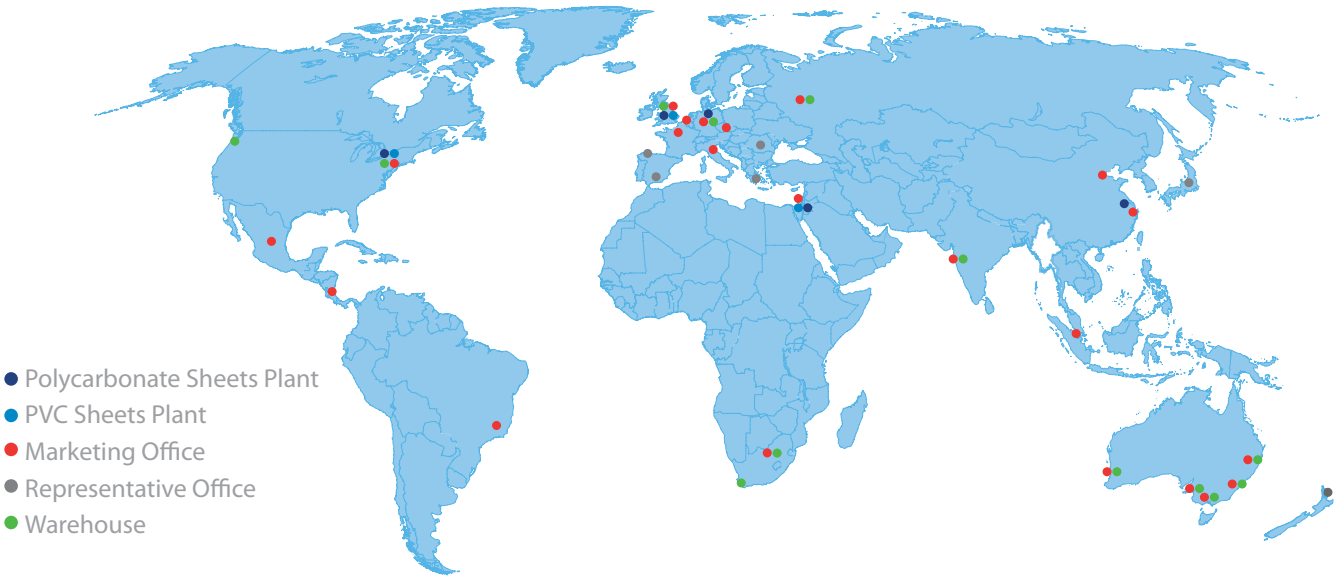
Palram has pioneered the production of extruded Polycarbonate & PVC Panels and has relied on innovative product culture since more than 50 years now . New technologies have been continuously adapted to refine the existing manufacturing methods. Palram also keeps track with market demands in order to create the most suitable panel for each application.

## Research & Development

As part of the tradition of producing high quality panels, Palram is constantly adapting new technologies to refine its manufacturing methods. The accumulated knowledge and production technology expands our product range and allows us to match our products more specifically to every project and application.

## Global Presence

Palram operates production plants in Europe, the Americas and on the shores of the Mediterranean, allowing constant logistic support.











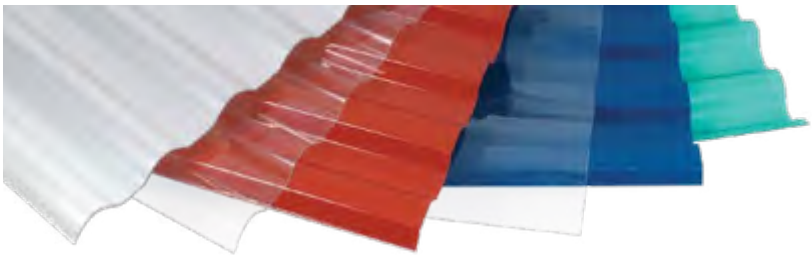




# PALRAM ACQUIRES TUFLITE

Palram India acquires TUFLITE POLYMERS LIMITED, a leading brand of Polycarbonate sheets across India. By incorporating TUFLITE® deep-rooted presence across India and customer base into Palram's growing portfolio, Palram looks forward to delivering exceptional products & value to a much larger market.

# SUNTUF®
## POLYCARBONATE CORRUGATED SHEET











### Overview

SUNTUF corrugated polycarbonate sheet is an outstanding roofing material that offers superior physical properties.
SUNTUF is favored by professionals for non-corrosive industrial structural roofing and siding, and by home owners for a multitude of DIY applications, due to its high impact strength and wide service temperature range.
The unlimited profile range and high light transmission allow SUNTUF to be integrate as high quality skylight with any roofing profile. SUNTUF is also available in a wide variety of colors and light diffusion options.

### Main Benefits

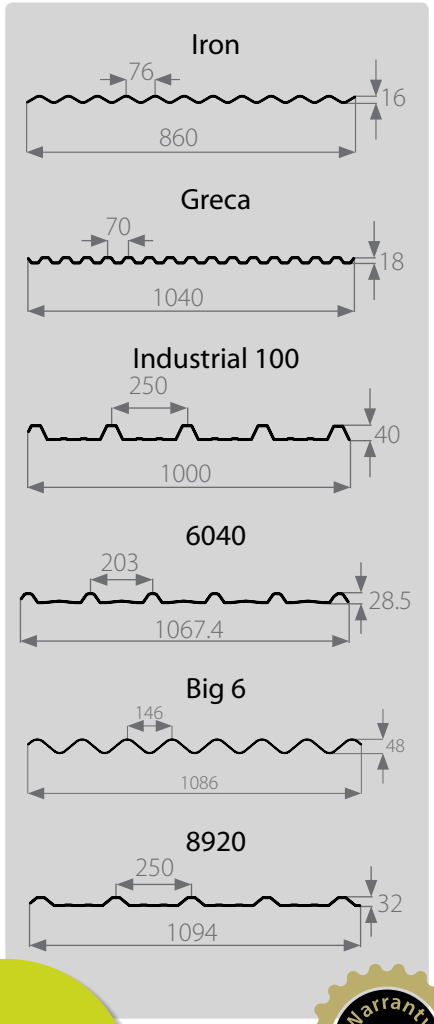
- Impact resistant: virtually unbreakable
- Transparent: up to 90% light transmission
- Weather and UV resistant
- Wide service temperature range
- UV blocking - better than sunscreens
- Lightweight
- Good flammability rating - does not emit toxic gasses when burning
- Warranted against yellowing, loss of light transmission and hail penetration
- Easy to work with and install

Metal Match Technology

### Colors

| Group | Colors | | | |
|---|---|---|---|---|
| Clear | Clear | | | |
| Transparent | Solar Gray | Bronze | Sky Blue | |
| | Transparent Green | Smart Gray | Breeze | |
| Translucent | White Opal | White dwwiffuser | Solar Control (Solar Metallic Grey) | |
| | Solar Ice | Mist Green | Smooth Cream | |
| Opaque | White | Green | Blue | Red Brick |

## Profiles




More then 400 Profiles

Warranty 10 Years

## Flammability

SUNTUF has a low flammability rating and doesnot emit toxic gasses when burning.

| Standard | *Classification |
|---|---|
| EN13501 | d0 ,B, s1 |
| 5 ,4 ,92501 NFP | 1-M |
| 4102 DIN | 1-B |
| 723 UL | 47.0 ,4.7 |
| Los Angeles Building Code | (2603 Section) CC2 |
| 9176 ,9177 UNI | 2 Class |

## We Match Any Profile!

### Matching New Profiles

- Matching any profile!
- No additional costs!
- Matching in no time!

### Profile ordering procedure:

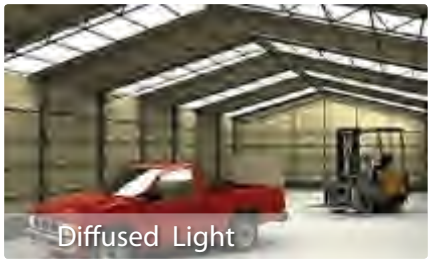
- Submit a detailed drawing or a full width sample.
- Palram will return a profile drawing for approval.



## SUNTUF® Rooflights

Direct VS. Diffuser Light Simulation




## Typical Physical Properties

| Property | *Method | Conditions | Units | Value |
|---|---|---|---|---|
| Density | D-792 | | g/cm³ | 1,2 |
| Heat deflection temperature (H.D.T) | D-648 | Load: 1.82 MPa | C° | 135 |
| Service Temperature Range (Short term) | | | C° | -50 to +120 |
| Service Temperature Range (Long term) | | | C° | -50 to +100 |
| Coefficient of linear thermal expansio | D-696 | | cm/cm°C | 6.5 x $10^{-5}$ |
| Thermal conductivity | C-177 | | W/m K | 0.21 |
| Tensile strength at yield | D-638 | 10 mm/min | MPa | 62 |
| Tensile strength at break | D-638 | 1 mm/min | MPa | 65 |
| Elongation at yield | D-638 | 10 mm/min | % | 6 |
| Elongation at break | D-638 | 10 mm/min | % | >70 |
| Tensile modulus of elasticity | D-638 | 1 mm/min | MPa | 2,300 |
| Flexural strength mechanica | D-790 | 1 mm/min | MPa | 93 |
| Flexural modulu | D-790 | 1.3 mm/min | MPa | 1,890 |
| Impact falling weight | (ISO 6603/1 E50) | 0.8 mm sheet | J | 50 |
| Rockwell hardness | D-785 | | R Scale | 118 |
| Light transmission | D-1003 | Clear sheet | % | 90% |
| Haze | D-1003 | Clear sheet | % | <0.5 |
| Yellowness index | D-1003 | Clear sheet | YI | <1 |

* ASTM method except where noted otherwise

# PALRUF®
## PVC CORRUGATED ROOFING & WALL SHEETS





### Overview

PALRUF corrugated PVC sheet is an all-round durable roofing solution, combining excellent resistance to chemicals, fire and impact.
PALRUF endures harsh chemical environments where other roofing materials will quickly become corrode, making it ideal for roofing, siding and cladding applications in industry, agriculture and construction.
PALRUF is available in a variety of clear and opaque colors and many profiles. It can be easily installed in DIY applications around the house using ordinary tools.



### Main Benefits

- Non Corrosive – Will not corrode or rust
- Resistant against widest range of Chemicals
- Durable & High Impact Strength
- UV & Weather Resistant
- Doesn't fad or yellow
- Excellent appearance
- Better Sound and Heat Insulation
- Meeting international flre standards
- Easily handled , Quickly installed

Heat Insulating

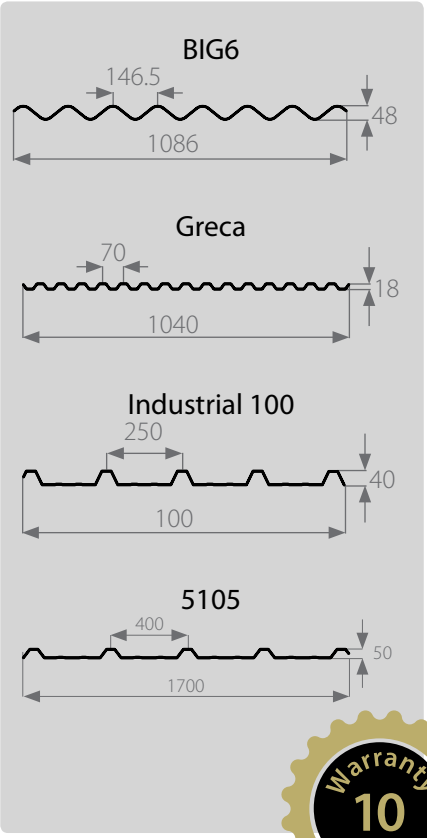
Chemical Resistant

Non Corrosive





## Profiles







## Load / Span Data - PALRUF® 100

| Load (kg/m²) | Maximum Roof Span (mm) | | | | | |
|---|---|---|---|---|---|---|
| | 2.0 mm | | 2.5 mm | | 3.0 mm | |
| | Mid Span | End Span | Mid Span | End Span | Mid Span | End Span |
| 50 | 1800 | 1350 | 1900 | 1425 | 2000 | 1500 |
| 75 | 1700 | 1275 | 1800 | 1350 | 1900 | 1425 |
| 100 | 1600 | 1200 | 1700 | 1275 | 1800 | 1350 |
| 125 | 1500 | 1125 | 1600 | 1200 | 1700 | 1275 |
| 150 | 1400 | 1050 | 1500 | 1125 | 1600 | 1200 |
| 175 | 1300 | 975 | 1400 | 1050 | 1500 | 1125 |
| 200 | 1200 | 900 | 1300 | 975 | 1400 | 1050 |

## Load / Span Data - PALRUF® 5150

| Load (kg/m²) | Maximum Roof Span (mm) | | | | | |
|---|---|---|---|---|---|---|
| | 2.0 mm | | 2.5 mm | | 3.0 mm | |
| | Mid Span | End Span | Mid Span | End Span | Mid Span | End Span |
| 75 | 2250 | 1750 | 2350 | 1800 | 2450 | 1850 |
| 100 | 2200 | 1700 | 2300 | 1750 | 2400 | 1800 |
| 125 | 2150 | 1650 | 2250 | 1700 | 2350 | 1750 |
| 150 | 2100 | 1600 | 2200 | 1650 | 2300 | 1700 |
| 175 | 2050 | 1550 | 2150 | 1600 | 2250 | 1650 |
| 200 | 2000 | 1500 | 2100 | 1550 | 2200 | 1600 |
| 200 | 1200 | 900 | 1300 | 975 | 1400 | 1050 |

## Flammability

PALRUF® meets most stringent flammability ratings.

| Standard | *Classification |
|---|---|
| DIN 4102 | B-1 |
| BS 476/7 | Class 1 |
| NSP 92501, 5 | M-1 |
| FMRC 4870 | Class 1 |



## Typical Physical Properties

| Property | *Method | Conditions | Units | Value |
|---|---|---|---|---|
| Density | D-792 | | g/cm³ | 1.4 |
| Heat deflection temperature (H.D.T) | D-648 | Load: 1.82 MPa | °C | 67-61 |
| Service Temperature Range | | | °C | -20 to +50 |
| Thermal conductivity | C-177 | | W/m K | 0.16 |
| Coefficient of linear thermal expansion | D-696 | | cm/cm°C | 6.3 x $10^{-5}$ |
| Impact strength | ISO 6603/1 | 2 mm sheet | J | 45 - 60 |
| Tensile strength at yield | D-638 | 10 mm/min | MPa | 50 - 66 |
| Tensile strength at break | D-638 | 10 mm/min | MPa | 39 - 53 |
| Elongation at yield | D-638 | 10 mm/min | % | 3 |
| Elongation at break | D-638 | 10 mm/min | % | >80 |
| Tensile modulus of elasticity | D-638 | 1 mm/min | MPa | 2,900 |
| Flexural strength | D-790 | 1.3 mm/min | MPa | 90 - 100 |
| Flexural modulus | D-790 | 1.3 mm/min | MPa | 2,700 |
| Rockwell hardness | D-785 | 0.8 mm sheet | R Scale | 105 - 115 |

* ASTM method except where noted otherwise

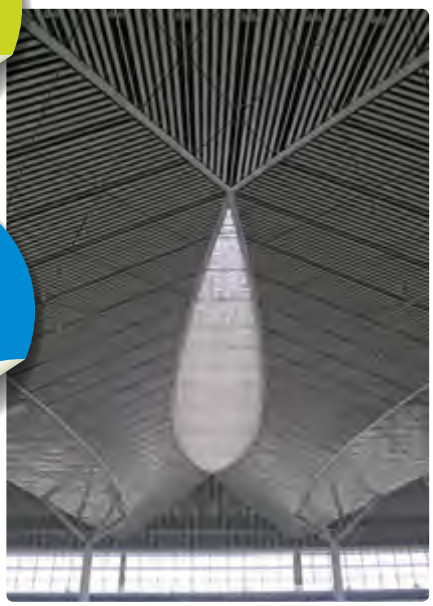
# SUNPAL®
## MULTIWALL MODULAR DAYLIGHT SYSTEM











### Overview
SUNPAL is an advanced multiwall polycarbonate panel system that combines proven design, light transmission, thermal insulation and strength. It offers a lightweight, leak-proof design that withstands very high loads and accommodates expansion and contraction. The system's distinct advantages make it ideal for long-term application on many types of projects. As with any true architectural glazing system, SUNPAL is appropriate for a variety of roofing and cladding designs, flat or curved.

### Main Benefits
- Withstands very high loads
- Accommodates expansion & contraction
- Simple & fast installation
- Leak-proof
- High thermal insulation
- Ideal for curved designs
- Double sided UV protection
- SolarSmart™ cool light colors

### Typical Applications
- Architectural roofing & glazing
- Commercial and retail roofing
- Sport facilities - translucent roofing
- Covered walkways, awnings & entrances
- Open markets - light roofing
- Service stations - translucent roofing
- Parking structure covering
- Swimming pool covers

100 % Leak Proof

Solar Smart - Heat Block

## Profiles

SUNPAL panels are of multiwall structure, available by thicknesses of 8mm, 10mm, 18mm and 20mm. Standard SUNPAL panels have UV protection on both sides, although panels with UV protection on one side only can be produced on special orders. Maximum panel length is 11.99m (typical stock length).

- 10/600

**Width:** 600mm
**Height:** 25.5mm
**Height with PC Joiner:** 33mm
**Weight:** 1.56 Kg/m, 2.60 Kg/m$^2$
**Min. cold bending radius:** 2.4m

- 18/1000

**Width:** 1000mm
**Height:** 33.5mm
**Height with PC Joiner:** 41mm
**Weight:** 3.11 Kg/m, 3.11 Kg/m$^2$
**Min. cold bending radius:** 3.0m

- 20/1000

**Width:** 1000mm
**Height:** 35.5mm
**Height with PC Joiner:** 43mm
**Weight:** 3.19 Kg/m, 3.19 Kg/m
**Min. cold bending radius:** 3.0m

Warranty 15 Years

Scan for Video

## Flammability

SUNPAL flammability classification appears in the attached table, based on a test performed by certified independent laboratories. The quoted certificate represents the flammability performance of the entire system.

| Standard | *Classification |
|---|---|
| EN 13501 | B, s1, d0 |

## Thermal Insulation

| Type | U-Value [Watts /m$^2$·°C] | R-Value [m$^2$·°C / Watt] |
|---|---|---|
| SUNPAL 10 mm | 2.10 | 0.47 |
| SUNPAL 18/20 mm | 1.50 | 0.67 |

## Typical Physical Properties

| Property | *Method | Conditions | Units | Value |
|---|---|---|---|---|
| Density | D-792 | | g/cm$^3$ | 1.2 |
| Heat deflection temperature (H.D.T) | D-648 | Load: 1.82 MPa | °C | 130 |
| Service Temperature - Short term | | | °C | -50 to +120 |
| Service Temperature - Long term | | | °C | -50 to +100 |
| Coefficient of linear thermal expansion | D-696 | | cm/cm°C | 6.5 x 10$^{-5}$ |
| Tensile strength at yield | D-638 | 10 mm/min | MPa | 62 |
| Elongation at break | D-638 | 10 mm/min | % | >80 |
| Impact falling dart | ISO 6603/1 | | J | 40-400 |
| Practical Thermal expansion/contraction rate | | | mm/m | 3 |

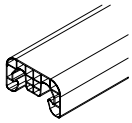
* ASTM method except where noted otherwise

## Colors

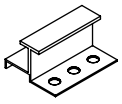
☐Clear, ■Bronze, ☐White Opal, ☐White Ice, ☐Solar Ice, ■Solar Control ■Grey, ■Green, ■Blue, ■Red And more.

## System Components

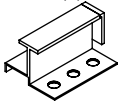
PC Joiner



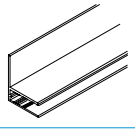
T-Fastener



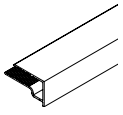
T-Stopper



Aluminum F-Profile



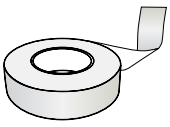
Aluminum Sealing Strip



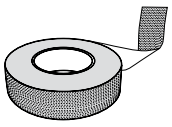
Metal Screw



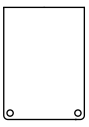
Aluminum Tape



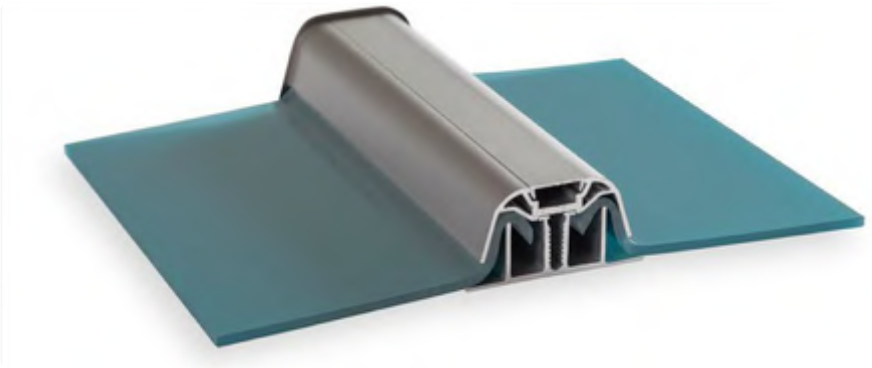
Breather Tape



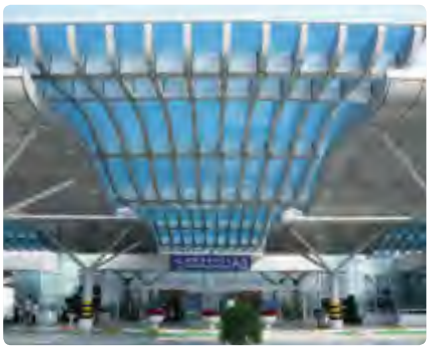
End-Cap



# SUNGLAZE®
## SOLID MODULAR DAYLIGHT SYSTEM











### Overview

Sunglaze is an architectural system that offers smart design, elegant appearance, versatility, low maintenance and sustainable performance to various architectural challenges. Sunglaze incorporates proprietary standing-seam profiling and glazing that enable wide spans and high loading capacity. It can be specified in various lengths to match different structures, including flat and curved designs. Sunglaze is easy to fasten; the panels are simply joined by an aluminum profile set that is enclosed at the ends by end-closures. Screws lock the system and fix it to the structure without any penetration through the panels. The Cap-plug completes the assembly, covering the screw head and provides a smooth appearance from above.

### Main Benefits

- Clear and elegant appearance
- Standing seam leak-proof performance
- Free thermal expansion
- Caulking and silicone free
- Withstanding high loads
- Easy, fast and safe installation
- Minimal maintenance

### Typical Applications

- Architectural projects
- Commercial and retail
- Sports venues roofing
- Covered walkways
- Open markets
- Service stations
- Entrances
- Pool covers

## Profiles

Sunglaze solid polycarbonate panels are offered in 3mm, 4mm and 6mm thickness. Panel width is determined by the system width, 600mm or 800mm. Maximum panel length is 11.9m (Typical stock length, equaling slightly more than 39 feet). Standard Sunglaze panels include UV protection on one side (UV protection on both sides is optional for special orders).

- 3/600
  **Width:** 584mm (600mm system)
  **Height:** 20mm
  **Weight:** 3.79 Kg/m$^2$, (2.20 Kg/m)
  **Min. cold bending radius:** 4m
- 4/600
  **Width:** 585mm (600mm system)
  **Height:** 21mm
  **Weight:** 5.05 Kg/m$^2$, (2.94 Kg/m)
  **Min. cold bending radius:** 4m
- 4/800
  **Width:** 785mm (800mm system)
  **Height:** 21mm
  **Weight:** 4.98 Kg/m$^2$, (3.90 Kg/m)
  **Min. cold bending radius:** 4m
- 6/800
  **Width:** 785mm (800mm system)
  **Height:** 21mm
  **Weight:** 7.453 Kg/m$^2$, (5.96 Kg/m)
  **Min. cold bending radius:** 4m

Scan for Video




## Typical Physical Properties

| Property | Method** | Conditions | Units | Value |
|---|---|---|---|---|
| Mechanical | | | | |
| Density | D-792 | | g/cm$^3$ | 1.2 |
| Tensile modulus of elasticity | D-638 | 1 mm/min | Mpa | 2,300 |
| Flexural strength | D-790 | 1 mm/min | Mpa | 93 |
| Flexural modulus | D-790 | 1.3 mm/min | Mpa | 2,600 |
| Notched impact strength Izod | D-256 | 23°C | J/m | 800 |
| Impact falling dart | ISO 6603/1d | 3mm | J | 158 |
| Impact - fall through | E-695 | | m/kg | 336 |
| Charpy Impact after Xenon Arc Exposure (D-6110) | D-2565-08 | 3000 hrs | % Loss of Impact Strength | <10 |
| Thermal | | | | |
| HDT (Heat Deflection Temperature) | D-648 | Load: 1.82Mpa | °C | 135 |
| Vicat softening temperature | D-1525 | Load: 1kg | °C | 150 |
| Service temperature - Short term | | | °C | -50 to 120 |
| Service temperature - Long term | | | °C | -50 to 100 |
| Coefficient of linear thermal expansion | D-696 | | cm/cm °C | 6.5 x 10$^{-5}$ |
| Thermal conductivity | C-177 | | W/m °K | 0.21 |
| Specific heat capacity | C-351 | | kJ/kg °K | 1.3 |
| Weathering | | | | |
| Color change | D-2244 | 60 months | ΔE | <3 |
| Yellowing index | E-313 | 60 months | Δ Yellowness Index | <10 |
| Light transmission | D-1003 | 10 years | % | <6 |
| Leakage / Structural | | | | |
| Water leakage | E-283 | 20 psf | | none |
| Air leakage | E-331 | 6.24 pfs (Condision of air leakage test) | cfm/ft$^2$ | 0.05 |
| Uniform load | E-330 | | psf | +140 / -45 |

*Properties in the table relate to the polycarbonate glazing panels in the Sunglaze system. **ASTM ethod except where noted otherwise. *** All tests results are pending sheet thickness

## Colors

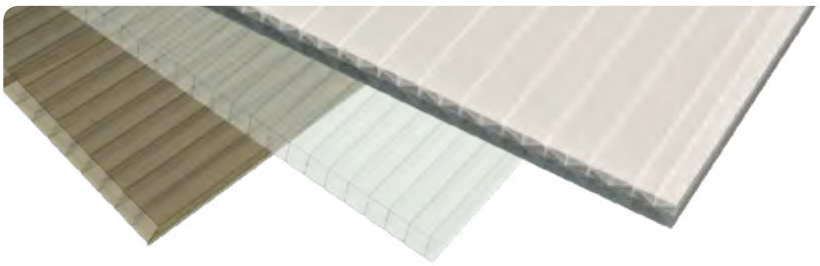
☐Clear, ■Bronze, ☐White Opal, ☐White Diffuser, ☐Solar Ice, ■Grey ■Solar Control, ■Solar Olympic, ■Smart Green, ■Smart Blue, ■Bluish Breez

## System Components

Base Cap Cap Plug End Closure Fixing Screw Locking Screw



# SUNLITE®
## MULTIWALL POLYCARBONATE SHEET





### Overview

SUNLITE's cellular polycarbonate structure yields a lightweight sheet with high impact strength and superior thermal insulation. High light transmission makes SUNLITE ideal for varied roofing, wall cladding, and glazing applications.
SUNLITE has a wide product range: from anti-condensation treatment for greenhouses and garden centers to SolarSmart™ sheets that create cool climatic conditions.
Interior designers and advertisers take advantage of SUNLITE's special appearance and add a unique touch to their designs.



### Main Benefits

- High thermal insulation
- Lightweight and impact resistant
- High light transmission
- Excellent structural durability
- Weather and UV resistance
- Blocks virtually all UV radiation
- Easy to handle and install
- High fire performance rating
-

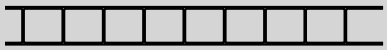
### Typical Applications

- Architectural roofing and glazing
- Skylights and sidelights
- Conservatories
- Covered walkways
- Displays, signage and decorations
- Industrial roofing and glazing
- Residential roofing and glazing
- Covered swimming pools
- Agricultural greenhouses



## Profiles

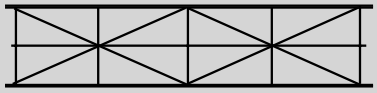
Twin Wall



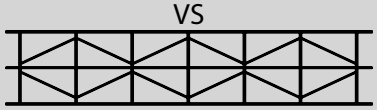
Triple wall



X- Lite



VS



**Length:**

11.8m

**Thickness:**

4-40mm

**Standard:**

Clear, Solar Grey, Bronze, Red, Blue, Green

**Translucent:**

Blue, Green

**Multi-Layered:**

Bronze/Opal, Solar Guard (Solar-control/Opal)




## Flammability

SUNTUF has a low flammability rating and doesnot emit toxic gasses when burning.

| Standard | *Classification |
|---|---|
| BS 476/7 | Class 1 |
| ASTM D-635 | CC-1 (SUNLITE® FR) |
| EN 13501 | B, s1, d0 |
| ASTM E-84 | Class A |

## Typical Physical Properties

| Property | *Method | Conditions | Units | Value |
|---|---|---|---|---|
| Density | | D-792 | g/cm$^3$ | 1.2 |
| Heat deflection temperature (H.D.T) | D-648 | Load: 1.82 MPa | °C | 135 |
| Service Temperature - Short term | | | °C | -20 to +50 |
| Service Temperature - Long term | | | °C | -50 to +100 |
| Coefficient of linear thermal expansion | D-696 | | cm/cm°C | 6.5x10$^{-5}$ |
| Tensile strength at yield | D-638 | 10 mm/min | MPa | 62 |
| Elongation at break | D-638 | 10 mm/min | % | >90 |
| Impact falling dart | ISO 6603/1 | | J | 40-400 |
| Practical thermal expansion/contraction | | | mm/m | 3 |

* ASTM method except where noted otherwise

## Product Range

**SUNLITE®** / UV protected on one side
**SUNLITE® UV2** / UV protected on both sides
**SUNLITE® Plus** / With anti-condensation
**SUNLITE® Solar Control** / Metallic reflective
**SUNLITE® CL** / Pearlescent reflective
**SUNLITE® SLT** / Translucent reflective
**SUNLITE® Smart** / Transparent reflective

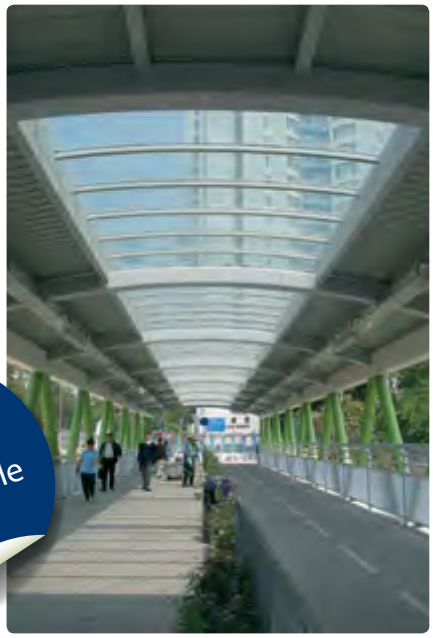
# PALSUN®
## FLAT SOLID POLYCARBONATE SHEET





### Overview

PALSUN combines superior impact resistance and clarity, making it the material of choice for many demanding applications. It is virtually unbreakable, yet it is transparent as glass at less than half its weight.
PALSUN can be cold bent and easily fabricated and formed, making it ideal for roofing, skylights, architectural roofing and glazing.
Clear, translucent and opaque PALSUN sheets are also ideal as machine guards, sound barriers, anti-vandal and safety glazing. PALSUN Diffuser and LB sheets are extensively used in light boxes.



### Main Benefits

- High impact resistance - Virtually unbreakable
- High clarity and light transmission
- Weather and UV resistant
- Wide service temperature range
- Blocks harmful UV radiation
- Good acoustical insulation
- Lightweight
- Easy to handle and install
- Versatile, formable, and machinable




### Typical Applications

- Transparent acoustic barriers
- Sport venues
- Skylights
- Shopping centers
- Covered walkways
- Sun rooms and conservatories
- Pergolas and covered patios
- Thermoformed, vacuum formed, bent and fabricated items








## Profiles

### Colors

**Transparent:** Clear, Solar Grey, Bronze, Red, Blue, Green, Breeze.
**Translucent:** Yellow, Red, White Opal, White Diffuser, Solar Control, Solar Ice, Solar Olympic.
**Opaque:** Dark Green, Red Brick, Black, Dark Blue, Cream, Light Grey, Dark Grey, Brown, Off White.

### Dimensions

**Width:**
1250x2050mm (49x81in.)
1220x2440mm (31x96in.),
2050x3050mm (81x120in.)

**Thickness:**
1-12mm / 0.059-0.472in.

### Product Range

**PALTUF™ /** General purpose (Indoor use)
**PALSUN /** UV protected on one side
**PALSUN® UV2 /** UV Protected on both sides
**PALSUN® FR /** Fire retardant
**PALSUN® Matte /** Anti-glare finish
**PALSUN® Solar Control /** Metallic reflective
**PALSUN® Breeze /** Transparent reflective
**PALSUN® LB /** Outdoor light-box applications




## Flammability

| | Standard | *Classification |
|---|---|---|
| **PALSUN®** | EN13501 | B, s1, d0 |
| | BS 476/7 | Class 1 |
| | NSP 92501, 4 | M2 |
| | UL Classified | V2 (File e221255) |
| | ASTM D-635 | CC1 |
| **PALSUN® FR** | UL 94 | V0 (File e221255) |

## PALSUN® VS Glass

| | PALSUN | Glass |
|---|---|---|
| Design Life | 40 Years | Untill breakage |
| Weight (Density) | Light (1.2 g/cm²) | Heavy (2.8 g/cm²) |
| Impact Resistance | Excellent (200 times more than glass) | Fair |
| Wind Resistance | Good | Good |
| Installation / Work Safety | Excellent (Safe - Virtually Unbreakable) | Poor (Dangerous) |
| % Light Transmission | 90% | 92% |
| Thermal Insulation | Good (Considerable better than glass) | Poor |

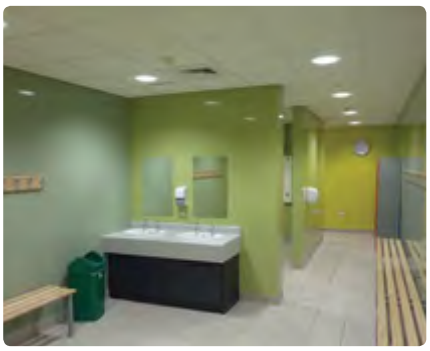
## Typical Physical Properties

| Property | *Method | Conditions | Units | Value |
|---|---|---|---|---|
| Density | D-792 | | g/cm³ | 1.2 |
| Heat Deflection Temperature | (D-648) | Load: 1.82 MPa | °C | 130 |
| Service Temperature - Short Term | | | °C | -50 to +120 |
| Service Temperature - Long Term | | | °C | -50 to +100 |
| Coefficient of linear thermal expansion | D-696 | | cm/cm°C | 0.065 |
| Thermal Conductivity | (C-177) | | W/m K | 0.21 |
| Tensile Strength at Yield | (D-638) | 10 mm/min | MPa | 65 |
| Tensile Strength at Break | (D-638) | 10 mm/min | MPa | 60 |
| Elongation at yield | D-638 | 10 mm/min | % | 6 |
| Elongation at break | D-638 | 10 mm/min | % | >90 |
| Tensile modulus of elasticity | D-638 | 1 mm/min | MPa | 2,300 |
| Flexural strength | D-790 | 1.3 mm/min | MPa | 100 |
| Flexural modulus | D-790 | 1.3 mm/min | MPa | 2,600 |
| Impact Falling Weight | (ISO 6603/1 E50) | 3mm sheet | J | 158 |
| Rockwell Hardness | (D-785) | | R Scale | 125R |
| Light Transmission | (D-1003) | | % | 90 |
| Haze | (D-1003) | | % | >0.5 |
| Yellowness Index | (D-313) | | YI | >1 |

* ASTM method except where noted otherwise

# PALCLAD™ Pro
## PVC WALL CLADDING SYSTEM











### Overview

PALCLAD Pro cladding system is a practical, cost efficient and comprehensive solution for maintaining a clean and safe environment.
Covering large areas is simplified with PALCLAD Pro, which have already helped achieve high resistance to fire and chemicals in hospitals, pharmaceutical plants, restaurants and other locations.
The system features flat opaque PVC panels, manufactured by Palram for more than 50 years, combine with custom profiles and welding rods in matching colors. A wide range of colors and surface textures enables design versatility for many purposes. The system's durability ensures many years of service that is unaffected by repetitive cleaning routines.

### Main Benefits

- Complete wall cladding solutions
- Profiles or welding rods options
- Creates high sanitation environment
- Withstands repeated cleaning cycles
- Versatile design options through many colors and surfaces
- High impact strength
- High fire rating
- High electrical and thermal insulation
- Non toxic
- PALCLAD™ Pro HYG: Active hygiene

Anti Microbial

### Typical Applications

- Public Facilities
- Food Industry
- Healthcare
- Agriculture

Easy & Quick Installation

## Profiles

### Colors

| | |
|---|---|
| STD White | Pistachio |
| Off White | Desert Sand |
| Mink | Ocean Blue |
| Grey Cream | Green Grape |
| Light Grey | Red |
| Grey | Black |

* Other colors are available on special order, subject to minimum quality.

### Dimensions

**Thickness:** 2mm, 2.5mm
**Widthx Length:** 1500 x 3000mm

### Product Range

**PALCLAD™ Pro White**
White panels & welding rods / profiles

**PALCLAD™ Pro Colored**
Colored panels & welding rods / profiles

**PALCLAD™ Pro HYG**
Active hygiene protection (Silver ions)

**PALCLAD™ Pro HYG**
Low Smoke - meeting s2 classification

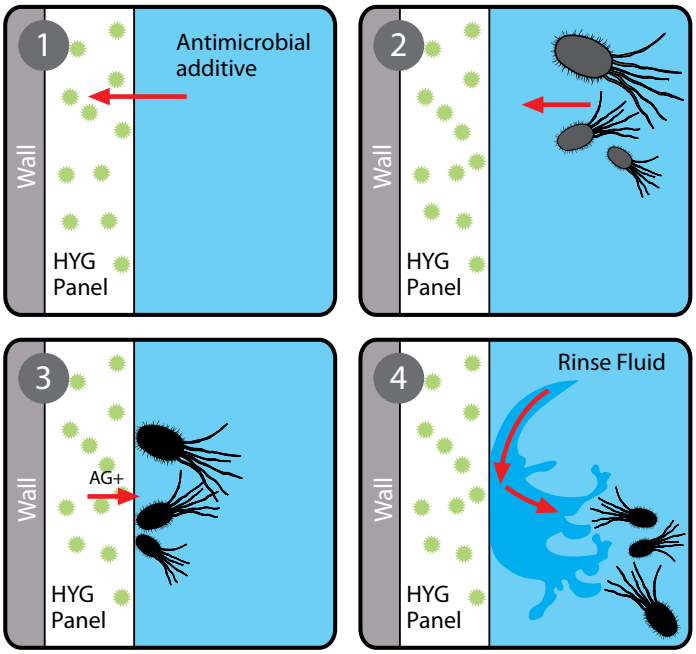



## Flammability

PALCLAD Pro is self-extinguishing and complies with the most demanding international fire resistance standards defined in the field of plastics, as shown in the attached table.

| | Standard | *Classification |
|---|---|---|
| **Standart Panel** | EN13501 | B, s3, d0 |
| | BS 476/7 | Class 0 |
| | ASTM E 84 | Class A |
| **LS Panel (Low Smoke)** | EN 13501 | B, s2, d0 |

## ?How Does Antimicrobial Action Work




**1.** PALCLAD® Pro HYG installed as wall cladding.

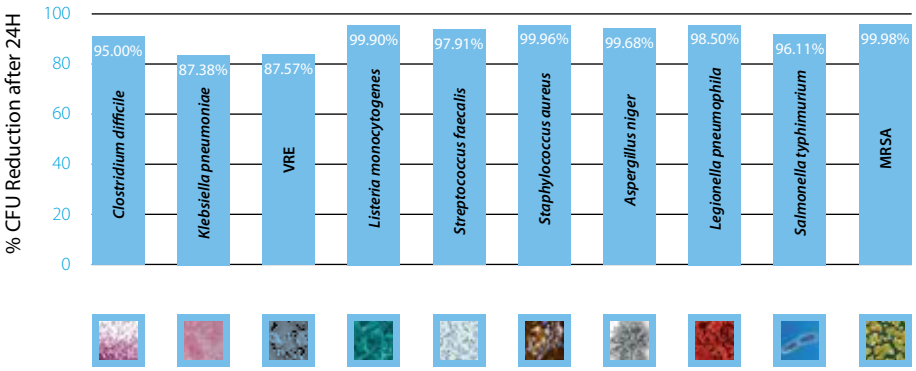
**2.** Microbes come in contact with the wall cladding while ions become active by moisture.

**3.** Ions attack microbes on the surface

**4.** Washing the wall with water or any other rinse fluide removes the dead microbes

## HYG Test - Microbe Elimination

10 common microbes were incubated on OALCLAD® Pro HYG for 24hrs at 37°c and >90% RH. The graphs shows HYG panels demonstrate high antimicrobial activity.



# Worldwide Projects



SUNPAL® - Zakho Stadium, Iraq



PALSUN® - Qingdao Railway Station China



SUNPAL® - School Basketball Court Taiwan



SUNLITE® - Roofing



SUNPAL® - School Basketball Court, Taiwan



SUNLITE® - Entrance Canopy



SUNPAL® - Hangzho Sport Park



SUNPAL® - Fun Republic Mall, India



SUNPAL® - Aquanation Center Melbourne, Australia



PALSUN® West Rail Hong Kong, China



SUNTUF® - Covered Walkway



SUNTUF® - Car Shed, Australia



PALRUF® - Pool Skylight Israel



PALRUF® - Chemical Plant



PALSUN® - Acoustc Barrier, Beijing China

PALRAM

TufLite


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